

YouTube Trending Analytics

An end-to-end data analysis project — from raw data to actionable business intelligence — built with Python, Pandas, and advanced statistical methods.

DATA SCIENCE PORTFOLIO

BUSINESS INTELLIGENCE

STATISTICAL ANALYSIS



Made with GAMMA

Technology Stack

Each tool was selected for a specific analytical role — together forming a complete, production-grade data workflow.



Python

Core language for data processing, analysis, and automation.



Pandas & NumPy

Data wrangling, aggregation, and numerical computation at scale.



Matplotlib & Seaborn

Static visualisations for EDA and publication-quality charts.



Plotly

Interactive dashboards, treemaps, and sunburst charts for stakeholder presentations.



SciPy

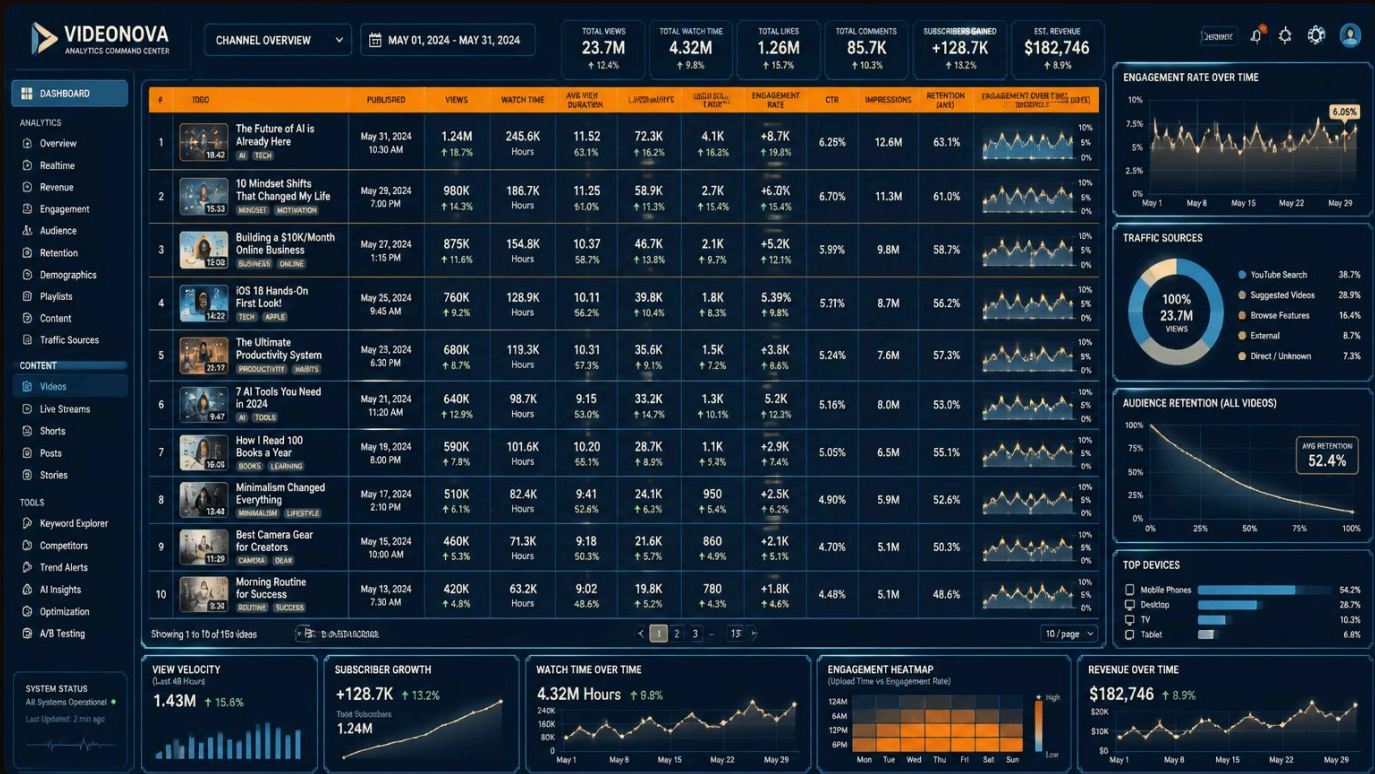
Hypothesis testing, correlation analysis, and statistical validation.



Jupyter Notebook

Reproducible, narrative-driven analysis combining code, output, and documentation.

Dataset Overview



The dataset captures **trending YouTube videos** across multiple countries, enriched with engagement metrics, temporal features, and content attributes – forming the foundation for a complete analytical pipeline.

Core Identity

video_id, title, description, channel_id, channel_title, tags

Engagement Metrics

view_count, like_count, comment_count, favourite_count

Temporal Features

publish_date, publish_hour, is_weekend, publish_quarter

Engineered Features

engagement_score, like_rate, duration_category, view_bucket

Data Cleaning Pipeline

Six sequential steps transformed raw, inconsistent data into a trustworthy analytical foundation. Each step addresses a distinct quality risk that could otherwise skew results or produce misleading business insights.

1

Missing Values

Imputed or removed incomplete records to prevent null propagation.

2

Duplicates

Eliminated repeated video entries that would inflate aggregate metrics.

3

Type Conversion

Cast dates, integers, and floats to correct dtypes for accurate computation.

4

Invalid Records

Filtered outliers and logically impossible values (e.g., negative durations).

5

Feature Creation

Derived engagement rates, duration categories, and temporal flags.

6

Validation

Final integrity checks confirmed a clean, analysis-ready dataset.

Feature Engineering

Raw columns were transformed into business-ready features that directly support strategic decision-making.

Engagement Score

Composite metric combining likes, comments, and views — enabling fair cross-category comparison.

Like & Comment Rate

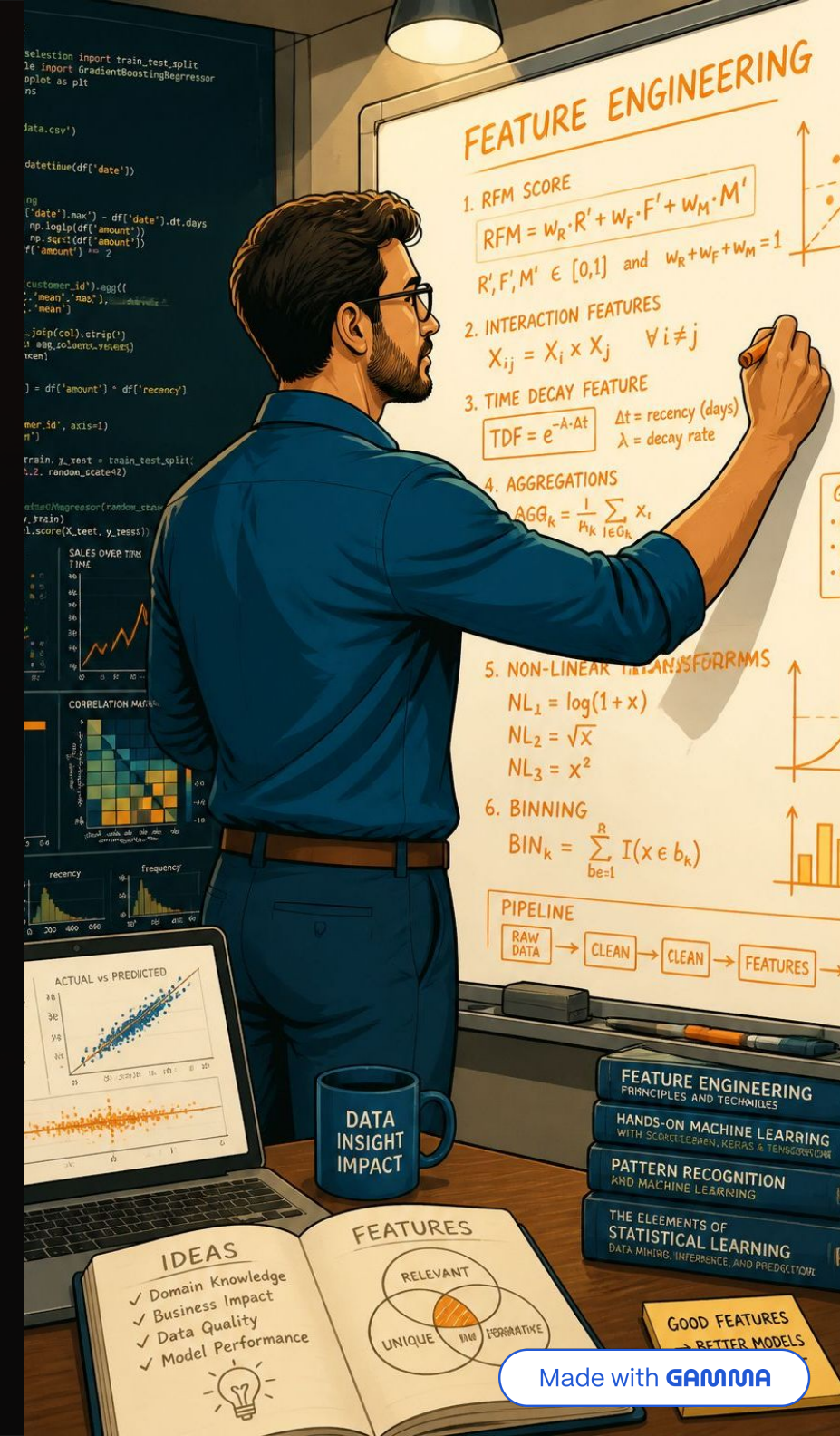
Normalised ratios revealing content quality independent of audience size.

Duration Category

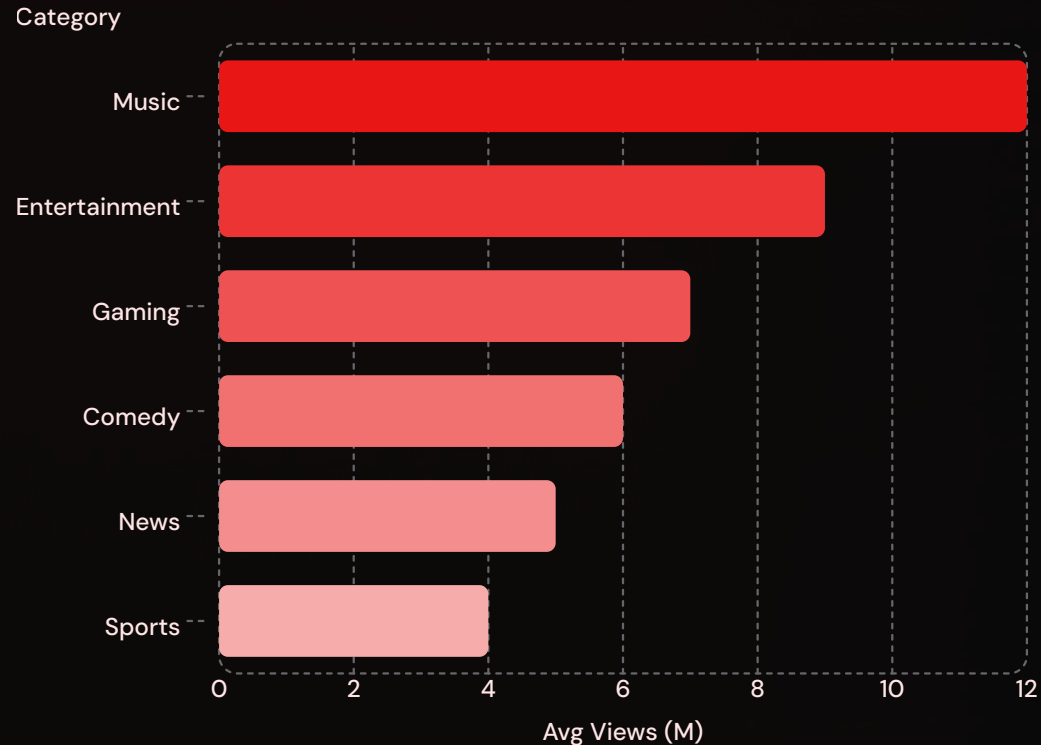
Binned videos into Short, Medium, and Long segments to identify optimal length per category.

Upload Session & Weekend Flag

Temporal flags uncovering peak publishing windows and weekend performance patterns.



Exploratory Data Analysis



Key EDA Findings

- **Music** dominates trending with the highest average views — a clear content opportunity.
- View distribution is **right-skewed**; a small fraction of videos drive the majority of traffic.
- Short-form content (under 5 min) shows the **highest engagement rates** across most categories.
- Peak upload hours cluster between **2 PM and 8 PM**, aligning with audience activity windows.

Business Questions – Key Insights

Twenty analytical questions were answered across four strategic themes, each driving a specific business recommendation.

Content Performance

Category, channel, and video-level analysis identifying top performers and content gaps.

Engagement Patterns

Like rates, comment rates, and engagement scores benchmarked by category and duration.

Upload Timing

Hour, session, day, month, and weekend analysis revealing optimal publishing windows.

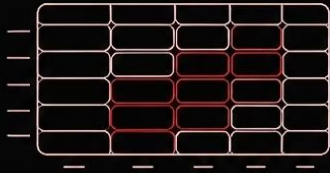
Creative Optimisation

Title length, tag count, and duration category tested against view and engagement performance.

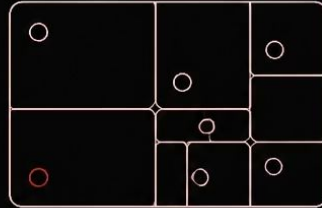


Advanced EDA & Time Series

Six advanced visualisation types used in the project:
Category Heatmaps showing engagement intensity across categories and time



Treemap displaying hierarchical view share by category and channel



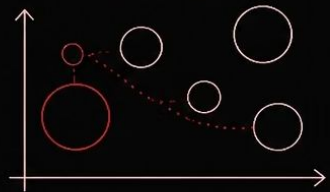
Sunburst revealing nested category-channel-video relationships



Radar Charts comparing multi-dimensional engagement metrics across categories



Bubble Charts mapping duration vs engagement vs views simultaneously



Correlation Matrix identifying statistical relationships between all numerical features



What the Advanced Analyses Revealed

- **Heatmaps** exposed strong category $\times$ time interaction effects on engagement.
- **Treemaps and Sunbursts** made hierarchical view-share intuitive for stakeholders.
- **Radar charts** enabled multi-metric category comparison in a single view.
- **Time series** uncovered clear seasonal upload patterns and weekly trending cycles.

❏ The Executive KPI Dashboard consolidated Total Videos, Views, Likes, Comments, Average Engagement, Categories, and Channels into a single stakeholder-ready summary.

Statistical Analysis

Rigorous hypothesis testing validated every insight — separating genuine patterns from random noise.



t-Test & Mann-Whitney U

Compared engagement between two groups (e.g., HD vs. non-HD videos).



ANOVA & Kruskal-Wallis

Tested for significant differences across multiple categories simultaneously.



Pearson & Spearman

Measured linear and monotonic relationships between duration, title length, and engagement.



Shapiro-Wilk & Levene

Confirmed normality and equal variance — prerequisites for parametric tests.



Chi-Square Test

Tested independence between categorical variables such as category and upload session.



Tukey HSD

Post-hoc pairwise comparisons following ANOVA to pinpoint exactly which categories differed.

Business Recommendations & Conclusion

Content Strategy

Prioritise Music and Entertainment; target 3–8 minute videos for peak engagement.

Upload Timing

Publish between 2 PM–8 PM on weekdays; avoid early-morning weekend uploads.

Optimisation

Enable HD and captions; craft titles of 50–70 characters with 8–12 targeted tags.

Future Scope

Extend to predictive modelling, real-time pipelines, and NLP-based sentiment analysis.

📌 **Project Impact:** This end-to-end analysis demonstrates production-ready skills in Python, Pandas, statistical testing, and business intelligence — portfolio-ready for Data Analyst, BI, and Data Scientist roles.

